

DNS SPESIFIC LOCALIZED THREAT INTELLIGENCE

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Project Proposal Report

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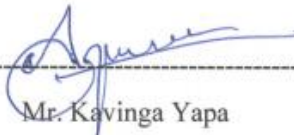
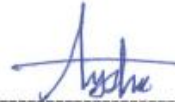
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DECLARATION

I hereby declare that this is my own work, and that this proposal has not incorporated, without proper acknowledgment, any material previously submitted for a degree or diploma at any other university or institution of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by any other person, except where explicit acknowledgment is made in the text.

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ABSTRACT

Traditional Threat Intelligence (TI) platforms often rely on large-scale, global data feeds that generate a significant amount of noise, making it difficult to identify relevant threats for specific sectors and regions. This project proposes a DNS Threat Intelligence (DNS-TI) system designed to overcome these limitations by integrating localized threat intelligence, AI-driven threat scoring, and dark web monitoring. By leveraging the Malware Information Sharing Platform (MISP) as the central platform for data storage, correlation, and sharing, this system aims to provide highly relevant and actionable DNS threat intelligence tailored to specific industries and regions.

The proposed system will incorporate proactive intelligence by monitoring dark web sources to detect threats before they materialize into full-scale attacks. An AI-based threat scoring mechanism will prioritize DNS-related threats based on context, risk, and source reliability, ensuring that only the most critical threats are highlighted for immediate action. Additionally, the system will enable feedback integration with DNS anomaly detection modules to continuously refine and improve threat intelligence.

Expected outcomes include a functional prototype capable of reducing irrelevant alerts, improving early detection, and providing actionable intelligence for DNS security teams. The system will also be evaluated based on accuracy, precision, recall, timeliness of detection, and its ability to reduce noise compared to traditional global TI feeds.

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1. INTRODUCTION

1.1. Background & Literature Survey

The Domain Name System (DNS) is a critical component of internet infrastructure, responsible for translating human-readable domain names into machine-readable IP addresses. Due to its fundamental role, DNS has become a primary target for various cyberattacks, including phishing, malware Command and Control (C2), data exfiltration through tunneling, and Domain Generation Algorithm (DGA)-based communication [1], [2]. Recent studies highlight the increasing use of DNS abuse by threat actors to bypass traditional security mechanisms such as firewalls and Intrusion Detection Systems (IDS), which are generally more effective at monitoring HTTP or email traffic [3].

DNS abuse is often overlooked by conventional security systems due to its seamless integration with regular network traffic, making it difficult to detect without specialized monitoring. Research suggests that, while DNS queries are vital for analyzing network behavior, they remain insufficiently monitored compared to other types of internet traffic [3]. This creates a significant gap in cybersecurity, particularly as attacks leveraging DNS grow in sophistication and frequency [3], [4].

To address these challenges, Threat Intelligence (TI) platforms have emerged, providing organizations with early warnings about malicious domains, IP addresses, and attack behaviors [5]. Open-source platforms like Malware Information Sharing Platform (MISP) are increasingly used for aggregating and sharing Indicators of Compromise (IoCs) across organizations, enhancing collaborative defense efforts. MISP offers standardized formats, comprehensive APIs, and correlation capabilities, making it a valuable resource for cybersecurity communities [6].

Despite their utility, current TI solutions, such as PhishTank, Spamhaus, and URLHaus, often suffer from significant limitations, particularly in the context of DNS-related threats. These platforms typically provide global IoCs that are not tailored to specific sectors or regions, which can lead to irrelevant data and inefficient alerting [7], [5]. For example, an organization in the finance sector might receive alerts about phishing campaigns targeting retail industries, making threat detection less effective.

A significant gap in current TI systems is the lack of proactive intelligence, which can detect threats before they materialize. Proactive techniques, such as dark web monitoring, allow for the early detection of attack vectors, including malicious domains, phishing kits, and discussions about new campaigns [8]. This approach enables organizations to take preventive action, mitigating threats before they reach the operational phase [9].

Furthermore, AI-powered threat scoring is still underutilized in most TI platforms. While some systems leverage machine learning for anomaly detection, they often lack comprehensive AI-driven models that integrate source reputation, contextual features (e.g., WHOIS, Passive DNS), and dark web signals. This limits the accuracy of threat prioritization, allowing high-risk threats to go unnoticed while low-risk ones generate unnecessary alerts [9], [8].

Finally, many TI platforms fail to offer DNS-specific contextualization. Most existing platforms provide generic IoCs, but DNS-related threats, such as DGA botnets, DNS tunneling, and DNS spoofing, are often underrepresented or generalized [3], [4]. A DNS-focused intelligence system is essential to track these threats, offering actionable intelligence that directly pertains to DNS infrastructure and helping organizations mitigate DNS-related risks [6].

In conclusion, the gaps identified in current research emphasize the need for a localized, proactive DNS-focused TI system that integrates AI-based scoring and dark web monitoring to detect and mitigate threats before they materialize.

1.2.Research Gap

While several open-source and commercial Threat Intelligence (TI) platforms are available, they often face significant limitations when addressing DNS-related threats [6]. These platforms primarily focus on global Indicators of Compromise (IoCs), which may not be relevant for specific sectors, regions, or DNS-related attacks. Furthermore, current TI solutions typically react to threats only after they have been detected, offering limited preventive value. Additionally, few platforms leverage AI-based threat scoring or dark web monitoring, which can help identify emerging threats before they materialize into full-scale attacks [8].

The key gaps in existing solutions are:

Noise and Irrelevance:

Most current TI platforms aggregate data globally without filtering for sector-specific or regional relevance. For instance, a financial institution might receive alerts about phishing campaigns targeting retail industries, leading to noise and irrelevant data. This lack of localization results in inefficient alerting and missed targeted threats.

Absence of Proactive Intelligence:

Traditional TI systems are reactive, focusing on indicators from attacks that have already occurred, making them less effective at preventing future incidents. Proactive intelligence, such as dark web monitoring, provides an opportunity to detect discussions or mentions of domains, services, or attack vectors before they lead to active threats. This early detection can be pivotal in mitigating threats before they materialize.

Minimal Integration of AI Threat Scoring:

AI-based solutions for threat scoring and classification are still limited in current TI platforms. While some systems use machine learning for pattern detection, they rarely offer a comprehensive AI-driven risk scoring model that combines source reliability, contextual enrichment (e.g., WHOIS, PDNS), and real-time data from dark web monitoring. The absence of such models leads to inaccurate prioritization, where high-risk threats may go unnoticed, and low-risk ones trigger unnecessary alerts.

Lack of DNS-Centric Contextualization:

Many TI systems provide general IoC coverage, but DNS-specific threats (e.g., domain generation algorithms, C2 servers, DNS tunneling) are often underrepresented or treated as part of broader categories. A DNS-focused system is essential to track and assess DNS-related threats and their associated risks, ensuring that security teams receive actionable intelligence that directly relates to their DNS infrastructure.

Limited Feedback Integration with Anomaly Detection:

Most existing TI platforms do not offer a feedback mechanism that allows for the continuous refinement of threat data. Feedback from DNS anomaly detectors or security monitoring systems can help fine-tune the risk scores of indicators, improving the accuracy and relevance of future intelligence. A dynamic feedback loop is critical to adapt to evolving threats in real-time.

Lack of SIEM Integration:

Current TI solutions often operate in isolation and do not integrate well with broader security infrastructures, such as Security Information and Event Management (SIEM) systems. Integration with SIEM is critical for centralizing threat data, correlating events from different security systems, and triggering real-time alerts. A seamless SIEM integration ensures that threat intelligence can be fed into existing security workflows for rapid response and incident management.

Table 1 : Research Gap Comparison

Feature	Existing TI Platforms	Proposed DNS-TI System
Noise and Irrelevance	Indicators are globally sourced, often irrelevant to specific sectors or regions.	Localized threat intelligence based on sector, geography, and assets.
Proactive Intelligence	Reactive: Threats are identified only after attacks have occurred.	Proactive: Dark web monitoring and AI-driven risk scoring identify threats before they materialize.
AI Integration	Minimal or no integration of AI for threat scoring or credibility assessment.	AI-based threat scoring, integrating source reputation, contextual

		features, and dark web signals.
DNS Focus	General IoCs are covered without specific focus on DNS-related threats.	DNS-centric TI system focused on enriching and prioritizing DNS threat indicators.
Feedback Integration	Limited integration with other anomaly detection systems.	Continuous feedback loop with anomaly detection components for refining intelligence and scoring.
SIEM Integration	Limited or no integration with Security Information and Event Management (SIEM) systems.	Seamless integration with SIEM for centralized alerting and event correlation, enhancing overall security posture.

1.3. Research Problem

Based on the gaps identified in existing literature, researches failed to address the localized DNS problem. The key research problem can be stated as follows:

“How can a localized DNS Threat Intelligence platform be designed to deliver sector- and region-specific intelligence, proactively detect DNS-related threats from dark web sources, and apply automated AI-based scoring to reduce noise while increasing actionable relevance for organizations?”

2. OBJECTIVES

2.1.Main Objective

To design and implement a localized DNS Threat Intelligence platform that integrates open-source feeds, dark web monitoring, and internal anomaly detections, enhanced with AI-based threat scoring, to provide sector- and region-specific intelligence that is more actionable and proactive than traditional global TI feeds.

2.2.Specific Objectives

- Deploy MISP as the Threat Intelligence backend to provide standardized storage, sharing, and correlation of DNS-related threat indicators.
- Ingest OSINT and dark web sources (e.g., OpenPhish, URLHaus, Ahmia/curated onion forums) into MISP for DNS-centric threat coverage.
- Develop a localization tagging framework (e.g., sector:finance, country:LK, brand keywords) to prioritize threats relevant to a given organization or industry.
- Integrate a threat scoring mechanism that combines AI classification, source reputation, and contextual enrichment to generate automated risk scores and severity levels for DNS indicators based on their risk, source reliability, and contextual relevance.
- Ingest data from other anomaly detection components and feed enriched intelligence back to them, improving their detection accuracy and enhancing the overall system's feedback loop.

3. METHODOLOGY

The methodology of this research component focuses on building a localized DNS Threat Intelligence (DNS-TI) platform that integrates open-source feeds, project-specific anomaly detections, and selected dark web sources. The platform will be powered by the Malware Information Sharing Platform (MISP), which provides a standardized framework for data storage, correlation, and dissemination. The methodology is divided into three phases. Data acquisition, processing and enrichment, and integration with feedback mechanisms.

3.1. System Diagram

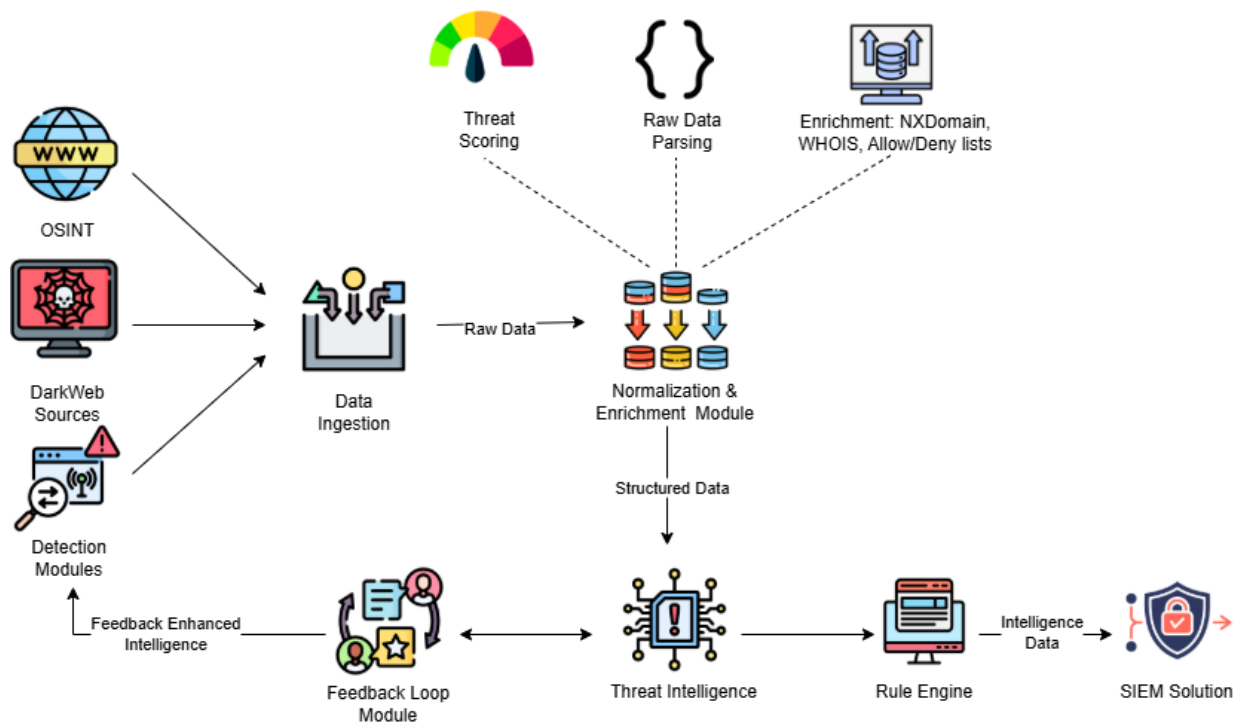


Figure 1 : System Diagram

3.2. Methodology Description

3.2.1. Data Acquisition

The first stage involves collecting DNS-related threat indicators from multiple complementary sources:

Table 2 : DNS Data Sources

Source Type	Example Feeds/Tools	Description	Purpose
Open-Source Intelligence (OSINT)	OpenPhish, PhishTank, URLHaus	Feeds that provide phishing and malware-related domains. These serve as the baseline dataset for known DNS threats.	Establish the foundation of known DNS threats to compare against.
Internal Anomaly Detectors	Custom Anomaly Detection Outputs	Outputs from other project components, such as DNS tunneling, data exfiltration, DGA-based C2 domains, and infrastructure abuse.	Reflects real-time anomalies observed in DNS traffic.
Dark Web Monitoring	Ahmia, Curated Onion Forums	Monitoring dark web forums and search services to identify mentions of domains, phishing kits, and DNS-related services.	Proactively gather intelligence on emerging threats before they surface.

3.2.2. Data Processing and Normalization

Once indicators are collected, they will be standardized and stored in MISP as events and attributes. A normalization pipeline will remove duplicates and align all feeds into a common schema. Each indicator will be tagged with contextual metadata such as sector relevance (e.g., finance, healthcare), geographic focus (e.g., Sri Lanka, South Asia), and organizational assets (e.g., brand names, registered domains). These

localization tags ensure that only relevant indicators are highlighted for the target environment.

3.2.3. Enrichment and Threat Scoring

Enrichment adds contextual value to raw indicators and is essential for prioritization.

The following DNS-specific features will be applied:

- WHOIS domain age and registrar analysis
- Passive DNS frequency and resolution patterns
- Lexical entropy and character analysis (for suspicious or algorithmic domains)
- Autonomous System Number (ASN) and GeoIP reputation checks
- DGA likelihood scoring

Based on these enriched features, a scoring model will assign each indicator a threat priority. The score will consider:

- Source reliability (OSINT, internal, dark web).
- Localization relevance (sector, region, or asset match).
- Enrichment evidence (WHOIS, PDNS, entropy, ASN).
- Optional AI credibility classification from dark web text.

Indicators that exceed a defined threshold will be promoted as high-priority events in MISP, while lower scores will be maintained in a watchlist for correlation.

3.2.4. Integration and Feedback Loop

The final stage is to establish a bi-directional feedback loop with the ther DNS anomaly detection components. Threat indicators generated by anomaly detectors will be ingested into the DNS-TI platform, enriched, and scored. At the same time, the scored and localized intelligence will be provided back to anomaly detectors, strengthening their detection logic. Confirmed results (true positives and false positives) from the detectors will then be used to refine scoring rules and thresholds in the DNS-TI system. This feedback mechanism ensures continuous improvement in the accuracy and precision of the intelligence produced.

3.2.5. Tools and Implementation

The following tools and technologies will be used in the implementation:

Table 3 : List of Tools

Tool/Technology	Description	Purpose in the System
MISP	Core platform for threat intelligence storage, correlation, and sharing.	Centralized storage and management of DNS threat indicators.
Python & PyMISP	Python programming language and PyMISP library for interacting with MISP and automating tasks.	Data ingestion, enrichment, automation, and MISP interaction.
Scikit-learn	A Python machine learning library for building models.	Used for implementing the AI-based threat scoring model (classification, risk scoring).
MySQL (MISP backend)	Database management system for storing MISP data.	Provides reliable and structured data storage.
Docker	Platform for developing, shipping, and running applications in containers.	Ensures easy deployment and environmental management.

3.2.6. Expected Outcome

The methodology will result in a prototype DNS Threat Intelligence platform that:

- **Prototype Implementation:** A fully functional DNS Threat Intelligence platform that integrates OSINT feeds, internal anomaly detection outputs, and dark web monitoring into MISP.
- **Localized Threat Intelligence:** A system that reduces irrelevant global indicators by applying localization techniques (sector-specific, region-specific) to prioritize only relevant threats for the organization.
- **AI-Driven Threat Scoring:** An automated threat scoring mechanism that uses AI classification, source reputation, and contextual enrichment to assign risk scores and severity levels to DNS indicators.

- **Proactive Threat Detection:** Incorporation of dark web monitoring to identify potential DNS threats before they become active, improving early detection capabilities.
- **Feedback Loop Integration:** A feedback-driven intelligence-sharing mechanism that allows DNS anomaly detection components to exchange enriched indicators and improve their detection models based on observed hits/misses.
- **Performance Evaluation:** The prototype's effectiveness will be evaluated through metrics such as:
 1. **Accuracy:** The proportion of correctly identified threats.
 2. **Precision and Recall:** Balancing false positives and false negatives in DNS threat detection.
 3. **Timeliness of Detection:** The speed with which new threats are identified and acted upon.
 4. **Reduction in Irrelevant Alerts:** A comparison between the prototype's relevance and the baseline of traditional global TI feeds.

3.3. Work Breakdown Structure

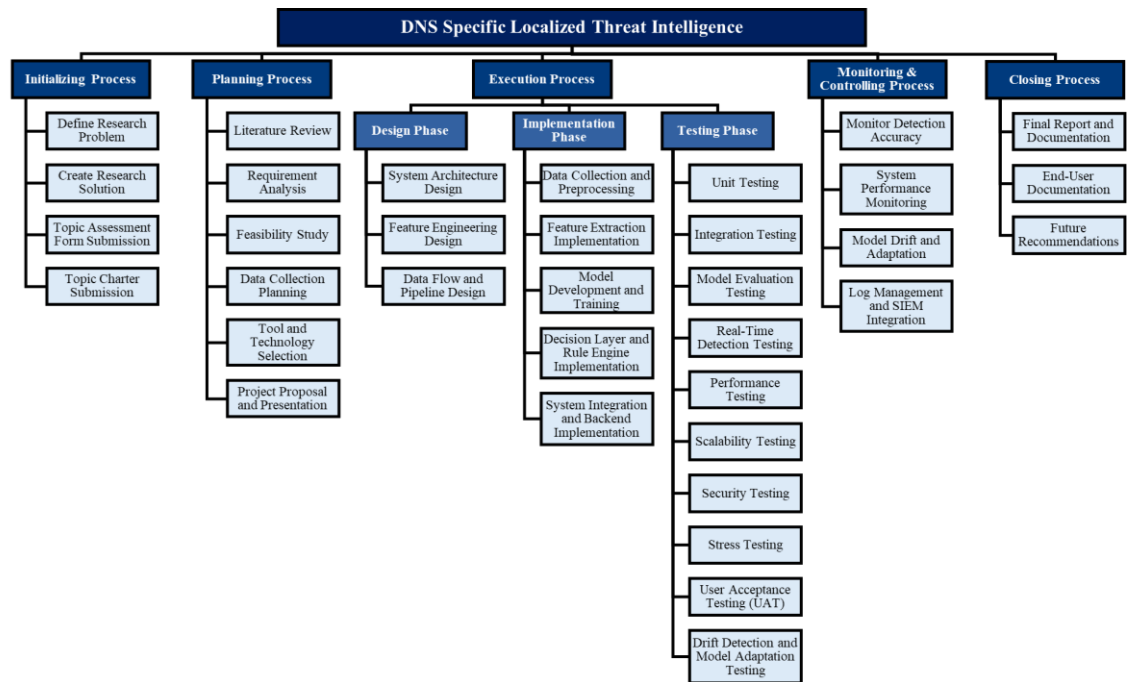


Figure 2 : Work Breakdown Structure

3.4. Gantt Chart

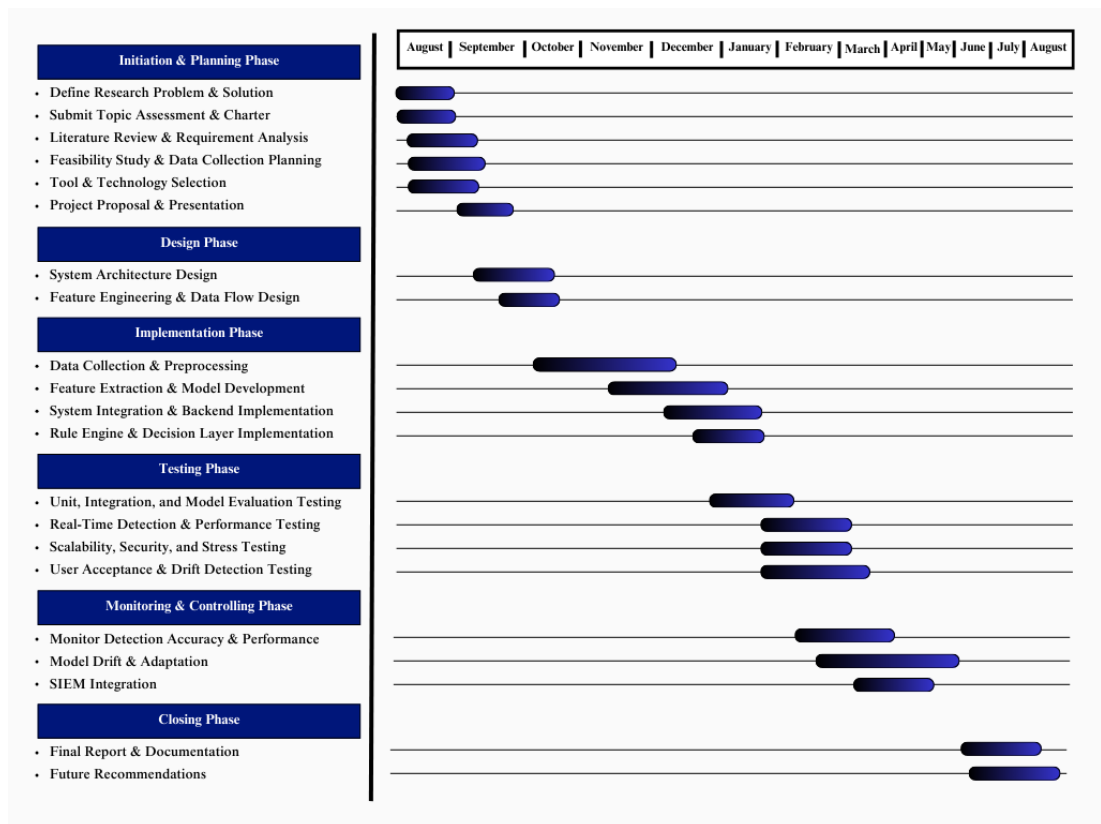


Figure 3 : Gantt Chart

4. PROJECT REQUIREMENTS

4.1.Functional Requirements

Data Ingestion:

- Collect DNS threat indicators from various sources, including OSINT (e.g., OpenPhish, URLHaus), internal anomaly detectors, and dark web monitoring tools (e.g., Ahmia, onion forums).
- Integrate these data sources into MISP for centralized storage and correlation.

Localization and Tagging:

- Implement a localization framework that tags threat indicators with sector (e.g., finance), region (e.g., Sri Lanka), and organizational assets (e.g., brand names).
- Prioritize and filter threats based on their relevance to the organization.

Threat Scoring and Severity Assessment:

- Develop an AI-based threat scoring mechanism that combines source reliability, contextual enrichment (e.g., WHOIS, PDNS), and risk scoring to assess the priority of DNS threats.
- Implement severity scoring to categorize threats based on their potential impact.

Feedback Loop Integration:

- Enable the exchange of threat intelligence between the DNS-TI platform and other project components, ensuring continuous refinement of threat indicators and scoring models based on feedback from anomaly detections.

Reporting and Alerting:

- Provide real-time alerts for high-priority threats, including actionable intelligence for further investigation.

- Generate reports for internal use or integration with other systems (e.g., SIEM, MISP).

4.2. Non-Functional Requirements

- **Performance:** The system must process incoming DNS threat indicators within minutes and ensure high performance when querying and retrieving intelligence from MISP and internal databases.
- **Scalability:** The system should scale to handle increasing data volumes and the addition of new data sources.
- **Reliability and Availability:** The system must maintain high uptime and have failover mechanisms for critical components like MISP and data ingestion pipelines.
- **Security:** All data, especially sensitive threat indicators, must be encrypted, with secure API communication and role-based access control.
- **Usability:** The user interface must be intuitive, and documentation should be clear for both users and administrators.

4.3. User Requirements

- **Security Analysts:** Should be able to view and query DNS threat intelligence, including enriched indicators and high-priority alerts, and interact with the threat scoring model.
- **System Administrators:** Must be able to configure and manage data feeds and monitor system performance.
- **Decision Makers:** Should receive summary reports highlighting key threats and trends relevant to their sector and region.

4.4. System Requirements

Table 4 : System Requirements

Category	Requirement	Description
Hardware	Multi-core Processor (Intel i7 or equivalent)	For handling large datasets and concurrent processing.
	At least 16GB of RAM	For smooth operation during high data loads.
	Sufficient disk space (100GB+)	For storing threat data, logs, and reports.
Software	MISP	For storing, correlating, and sharing threat intelligence.
	Python 3.x (PyMISP, Scikit-learn)	For data ingestion, threat scoring, and machine learning.
	Docker	For deploying components in isolated environments.
	MySQL or PostgreSQL	For backend data storage
Network	Internet access	For pulling OSINT feeds and monitoring dark web sources.
	Secure internal network	For safe communication between system components.

4.5. Use Cases

Use Case 1: Data Ingestion from OSINT and Dark Web Sources

- **Actor:** System Administrator
- **Goal:** To ingest threat intelligence data from OSINT feeds and dark web monitoring sources into the DNS Threat Intelligence system.
- **High-Level Steps:**
 1. The system administrator configures the OSINT feed sources (e.g., OpenPhish, URLHaus) and dark web monitoring tools (e.g., Ahmia).
 2. Data from these sources is automatically ingested into the MISP platform.
 3. The system validates and normalizes the data, ensuring compatibility with the existing threat intelligence format.

4. Data is tagged with appropriate localization labels (e.g., sector:finance, country:LK).
5. The system stores and correlates the threat indicators within MISP for further processing.

Use Case 2: Automated Threat Scoring and Risk Assessment

- **Actor:** Security Analyst
- **Goal:** To automatically generate threat scores and severity levels for DNS-related threats based on AI classification and contextual enrichment.
- **High-Level Steps:**
 1. The security analyst accesses the DNS Threat Intelligence platform and queries the latest DNS threat data.
 2. The system processes the data, applying threat scoring algorithms that incorporate AI classification, source reputation, and enrichment features (e.g., WHOIS, PDNS).
 3. The system calculates a risk score and assigns a severity level based on the threat's potential impact and relevance.
 4. High-priority threats are flagged for immediate action, and alerts are generated.
 5. The security analyst reviews the results and takes appropriate actions based on the generated insights.

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